

2020; Benzell et al., 2021). The realization of general purpose technologies' full potential requires extensive co-invention (Bresnahan and Trajtenberg, 1995; Bresnahan et al., 1996, 2002; Lipsey et al., 2005; Dixon et al., 2021), a costly and time-consuming process involving the discovery of new business procedures (David, 1990; Bresnahan, 1999; Frey, 2019; Brynjolfsson et al., 2021; Feigenbaum and Gross, 2021). Consequently, many studies of machine learning technologies focus on systems-level adoption, arguing that organizational systems may require redesign to effectively take advantage of novel machine learning advancements (Bresnahan, 2019; Agrawal et al., 2021; Goldfarb et al., 2023). Appropriately designed systems can yield considerable business value and improve firm performance (Rock, 2019; Babina et al., 2021; Zolas et al., 2021), with AI tools facilitating the discovery process (Cockburn et al., 2018; Cheng et al., 2022). By employing task-level information to assess whether LLMs fulfill the criteria of a general purpose technology, we seek to merge the two perspectives for understanding the technology-labor relationship.

We attempt to build on these diverse literature streams in several ways. Echoing (Felten et al., 2023), we focus our analysis on the impact of LLMs, rather than addressing machine learning or automation technologies more broadly. Additionally, we propose a novel method that employs LLMs, specifically GPT-4, to assess tasks for exposure and automation potential, thereby bolstering human scoring efforts. Subsequently, we aggregate our findings to occupations and industries, capturing the overall potential exposure in the contemporary U.S. labor market.

3 Methods and Data Collection

3.1 Data on Activities and Tasks Performed by Occupation in the US

We use the O*NET 27.2 database (O*NET, 2023), which contains information on 1,016 occupations, including their respective Detailed Work Activities (DWAs) and tasks. A DWA is a comprehensive action that is part of completing task, such as "Study scripts to determine project requirements." A task, on the other hand, is an occupation-specific unit of work that may be associated with zero, one, or multiple DWAs. We offer a sample of tasks and DWAs in Table 1. The two datasets we use consist of:

- 19,265 tasks, consisting of a "task description" and a corresponding occupation, and
- 2,087 DWAs, where most DWAs are connected to one or more tasks, and tasks may be associated with one or more DWAs, though some tasks lack any associated DWAs.

3.2 Data on Wages, Employment, and Demographics

We obtain employment and wage data from the 2020 and 2021 Occupational Employment series provided by the Bureau of Labor Statistics. This dataset encompasses occupational titles, the number of workers in each occupation, and occupation-level employment projections for 2031, typical education required for entry in an occupation and on-the-job training required to attain competency in an occupation (BLS, 2022). We use the BLS-recommended crosswalk to O*NET (BLS, 2023b) to link the O*NET task and DWA dataset and the BLS Labor Force Demographics (BLS, 2023a), which is derived from the Current Population Survey (CPS). Both of these data sources are collected by the U.S. government and primarily capture workers who are not self-employed, are documented, and are working in the so-called formal economy.

3.3 Exposure

We present our results based on an exposure rubric, in which we define **exposure** as a measure of whether access to an LLM or LLM-powered system would reduce the time required for a human to perform a specific

6.2 Implications for US Public Policy

The introduction of automation technologies, including LLMs, has previously been linked to heightened economic disparity and labor disruption, which may give rise to adverse downstream effects (Acemoglu and Restrepo, 2022a; Acemoglu, 2002; Moll et al., 2021; Klinova and Korinek, 2021; Weidinger et al., 2021, 2022). Our results examining worker exposure in the United States underscore the need for societal and policy preparedness to the potential economic disruption posed by LLMs and the complementary technologies that they spawn. While it is outside the scope of this paper to recommend specific policy prescriptions to smooth the transition to an economy with increasingly widespread LLM adoption, prior work such as (Autor et al., 2022b) has articulated several important directions for US policy related to education, worker training, reforms to safety net programs, and more.

6.3 Limitations and Future Work

In addition to those discussed above, we highlight some particular limitations of this work that warrant further investigation. Primarily, our focus on the United States restricts the generalizability of our findings to other nations where the adoption and impact of generative models may differ due to factors such as industrial organization, technological infrastructure, regulatory frameworks, linguistic diversity, and cultural contexts. We hope to address this limitation by extending the study’s scope and by sharing our methods so other researchers can build on them.

Subsequent research efforts should consider two additional studies: one exploring LLM adoption patterns across various sectors and occupations, and another scrutinizing the actual capabilities and limitations of state-of-the-art models in relation to worker activities beyond the scope of our exposure scores. For example, despite recent advances in multimodal capabilities with GPT-4, we did not consider vision capabilities in the α ratings on direct LLMs-exposure (OpenAI, 2023b). Future work should consider the impact of such capability advances as they unfold. Furthermore, we acknowledge that there may be discrepancies between theoretical and practical performance, particularly in complex, open-ended, and domain-specific tasks.

7 Conclusion

In conclusion, this study offers an examination of the potential impact of LLMs on various occupations and industries within the U.S. economy. By applying a new rubric for understanding LLM capabilities and their potential effects on jobs, we have observed that most occupations exhibit some degree of exposure to LLMs, with higher-wage occupations generally presenting more tasks with high exposure. Our analysis indicates that approximately 19% of jobs have at least 50% of their tasks exposed to LLMs when considering both current model capabilities and anticipated LLM-powered software.

Our research aims to highlight the general-purpose potential of LLMs and their possible implications for US workers. Previous literature demonstrates the impressive improvements of LLMs to date (see 2.1). Our findings confirm the hypothesis that these technologies can have pervasive impacts across a wide swath of occupations in the US, and that additional advancements supported by LLMs, mainly through software and digital tools, can have significant effects on a range of economic activities. However, while the technical capacity for LLMs to make human labor more efficient appears evident, it is important to recognize that social, economic, regulatory, and other factors will influence actual labor productivity outcomes. As capabilities continue to evolve, the impact of LLMs on the economy will likely persist and increase, posing challenges for policymakers in predicting and regulating their trajectory.

Further research is necessary to explore the broader implications of LLM advancements, including their potential to augment or displace human labor, their impact on job quality, impacts on inequality, skill development, and numerous other outcomes. By seeking to understand the capabilities and potential effects